1. Evaluating Real-Time Emotional Responses Using Bullet Screen Sentiment Analysis: Evidence from Electrodermal Activity

Accession number: 20245317605126
Authors: Xu, Zhao (1); Li, Qingchuan (1); Song, Yao (2, 3)
Author affiliation: (1) School of Humanities and Social Sciences, Harbin Institute of Technology, Shenzhen, China; (2) College of Literature and Journalism, Sichuan University, Chengdu, China; (3) Digital Convergence Laboratory of Chinese Cultural Inheritance and Global Communication, Sichuan University, Chengdu, China
Corresponding author: Li, Qingchuan(liqingchuan@hit.edu.cn)
Source title: Lecture Notes in Computer Science
Abbreviated source title: Lect. Notes Comput. Sci.
Volume: 15375 LNCS
Part number: 2 of 7
Issue title: HCI International 2024 – Late Breaking Papers - 26th International Conference on Human-Computer Interaction, HCII 2024, Proceedings
Issue date: 2025
Publication year: 2025
Pages: 240-253
Language: English
ISSN: 03029743
E-ISSN: 16113349
ISBN-13: 9783031768057
Document type: Conference article (CA)
Conference name: 26th International Conference on Human-Computer Interaction, HCII 2024
Conference date: June 29, 2024 - July 4, 2024
Conference location: Washington, DC, United states
Conference code: 313029
Publisher: Springer Science and Business Media Deutschland GmbH
Abstract: Bullet screens are attracting increasing attention as a way to express emotions and interact on short video platforms. Prior studies have used natural language processing (NLP) to analyze bullet screen sentiment in order to evaluate public opinion trends regarding a specific topic, movie, or product. However, few studies have investigated the effectiveness of using bullet screen sentiment analysis to predict real-time emotional responses. Thus, this study examined whether and to what extent bullet screen sentiment analysis can be used to evaluate and predict real-time emotional responses to videos by employing physiological electrodermal activity (EDA) measurements. A behavioral experiment was conducted in which eight college students wore a set of wireless galvanic skin sensors while watching three music videos (MVs) in random or-der. The participants' EDA data, including skin conductance responses and peak amplitudes, were then analyzed. Meanwhile, the sentiments expressed in the bullet screen comments on the three MVs were analyzed using three dictionary-based sentiment analysis algorithms: SnowNLP, BosonNLP, and Hel-loNLP. The bullet screen sentiment analysis and physiological measurement results were then compared using descriptive and correlation analyses. The bullet screen sentiment parameters were found to significantly correlate with the EDA measurements. This study confirms the effectiveness of using bullet screen sentiment analysis to predict participants' real-time emotional responses, providing a convenient and flexible way for enterprises and governments to detect public opinion trends and take action accordingly. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.
Number of references: 26
Main heading: Economic and social effects
Controlled terms: Physiological models - Video analysis
Uncontrolled terms: Activity measurements - Bullet screen - Electrodermal activity - Emotional response - Express emotions - Music video - Public opinions - Real- time - Sentiment analysis - Video-platforms
Classification code: 101.1 Biomedical Engineering - 971 Social Sciences - 1106.3.1 Image Processing
DOI: 10.1007/978-3-031-76806-4_18
Funding Details: Number: 62207008, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: -, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: GXWD20231129154726002, Acronym: -, Sponsor: -;
Funding text: This study was funded by the National Natural Science Foundation of China [grant number 62207008] and General Program of Stable Support Plan for Universities in Shenzhen [grant number GXWD20231129154726002].
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

2. Quantitative Research on Group Polarization of Bullet-screen Video Websites Based on BERTopic and XGBoost models (*Open Access*)

Accession number: 20254619519336
Authors: Dai, Jianhua (1); Wang, Yuran (1)
Author affiliation: (1) Nanjing University of Science and Technology, Jiangsu, Nanjing, China
Corresponding author: Wang, Yuran(1098162169@qq.com)
Source title: Proceedings of 2025 International Conference on Management Science and Computer Engineering, MSCE 2025
Abbreviated source title: Proc. Int. Conf. Manage. Sci. Comput. Eng., MSCE
Part number: 1 of 1
Issue title: Proceedings of 2025 International Conference on Management Science and Computer Engineering, MSCE 2025
Issue date: September 23, 2025
Publication year: 2025
Pages: 242-247
Language: English
ISBN-13: 9798400715969
Document type: Conference article (CA)
Conference name: 2025 International Conference on Management Science and Computer Engineering, MSCE 2025
Conference date: June 6, 2025 - June 8, 2025
Conference location: Dalian, China
Conference code: 213063
Publisher: Association for Computing Machinery, Inc
Abstract: The high interactivity feature of bullet-screen video websites has exacerbated the formation of group polarization. Effective analysis of the evolution law of opinions on bullet-screen video websites can help us grasp the public opinion disputes of events more clearly, and thereby provide strong support for public opinion management. Firstly, the BERTopic model is introduced to conduct topic modeling and sentiment calculation for the public opinion related to the event. Combined with the characteristics of bullet-screen video websites, a targeted quantification algorithm for sentiment divergence degree is designed. The XGBoost model is used to fit the sentiment score of video comments with the source prestige, and based on this, the sentiment divergence degree of the event and its sub-topics is calculated. Further carry out the research on the phenomenon of group polarization. Taking the relevant discussions about Chang'e-6 on the Bilibili website as an example for empirical research, the results show that introducing the degree of emotional divergence can more accurately judge the phenomenon of group polarization and effectively monitor the points of public opinion disputes.
© 2025 Copyright held by the owner/author(s)
Number of references: 11
Main heading: Social aspects
Controlled terms: Behavioral research - Polarization - Sentiment analysis
Uncontrolled terms: Bertopic - Bullet screen video website - Divergence degrees - Effective analysis - Evolution law - Group polarization - Interactivity features - Public opinions - Quantitative research - Sentiment divergence
Classification code: 101.5 Ergonomics and Human Factors Engineering - 901.4 Impact of Technology on Society - 971 Social Sciences - 1101 Artificial Intelligence - 1106.7 Computational Linguistics
DOI: 10.1145/3760023.3760062
Compendex references: YES
Open Access type(s): All Open Access, Gold
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

3. Impacts of Bullet-Screen Content and Behavioral Characteristics on Consumer Purchasing Behavior in Live Streaming E-Commerce: Moderating Effect of Broadcaster-Product Relevance

Accession number: 20251418184925
Title of translation: 直播电商弹幕内容与行为特征对消费者购买行为的影响研究——主播与产品关联度的调节效应
Authors: Ting, Chen (1); Honghao, Ding (1); Haoyu, Zhou (1); Jiang, Wu (1, 2, 3)
Author affiliation: (1) School of Information Management, Wuhan University, Wuhan; 430072, China; (2) The Center for Studies of Information Resources, Wuhan University, Wuhan; 430072, China; (3) Center for E-commerce Research and Development, Wuhan University, Wuhan; 430072, China
Corresponding author: Jiang, Wu(jiangw@whu.edu.cn)
Source title: Data Analysis and Knowledge Discovery
Abbreviated source title: Data. Anal. Knowl. Discov.
Volume: 9
Issue: 2

Issue date: February 25, 2025
Publication year: 2025
Pages: 159-171
Language: Chinese
ISSN: 20963467
Document type: Journal article (JA)
Publisher: Chinese Academy of Sciences
Abstract: [Objective] This study explores the impacts of bullet-screen(danmu)content and behavioral characteristics on consumers purchasing behavior in live-streaming e-commerce, as well as the moderating effect of host-product relevance. [Methods] First, we retrieved the bullet-screen data from the Douyin platform and the consumer data from the Huitun platform based on the Elaboration Likelihood Model. Then, we studied the impacts of bullet-screen content characteristics (central route) and behavior characteristics (peripheral route) on consumer purchasing behavior with text mining and zero-inflated negative binomial regression. We also discussed the moderating effect of host-product relevance with grouping regression. [Results] Information richness, social interaction degree and number of bullet-screen comments positively impact purchasing behavior. The emotional polarity of bullet screen comments exhibits an inverted U-shaped effect on purchasing behavior. Compared with live streaming rooms with low host-product relevance, those with high host-product relevance have broader positive impacts on purchase behavior. [Limitations] We only investigated the bullet-screen data from a single live-streaming e-commerce platform. [Conclusions] This study examines the factors influencing consumers' actual purchasing behavior from the perspective of bullet-screen comments. It provides insights for improving communication between merchants and consumers in live-streaming e-commerce, ultimately enhancing sales performance. © 2025 Chinese Academy of Sciences. All rights reserved.
Number of references: 47
Main heading: Sales
Controlled terms: Economic and social effects - Regression analysis
Uncontrolled terms: Broadcaster-product relevance - Bullet-screen characteristic - Consumer purchasing behaviors - E- commerces - Elaboration likelihood models - Live streaming - Live streaming E-commerce - Moderating effect - Purchase behavior - Purchasing behaviors
Classification code: 911.4 Marketing - 971 Social Sciences - 1202.2 Mathematical Statistics
DOI: 10.11925/infotech.2096-3467.2023.1424
Funding Details: Number: 72232006, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: -, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: 20JZD024, Acronym: MOE, Sponsor: Ministry of Education of the People's Republic of China; Number: -, Acronym: MOE, Sponsor: Ministry of Education of the People's Republic of China;
Funding text: The work is supported by the National Natural Science Foundation of China (Grant No. 72232006), Key Projects of Philosophy and Social Sciences Research,Ministry of Education (Grant No. 20JZD024).
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

4. A Study on Semantic Shift of Uncivilized Words in Bullet-Screen Comments

Accession number: 20241215755030
Authors: Fan, Juanjuan (1); Yang, Jianguo (2)
Author affiliation: (1) Training Department, Beijing Chinese Language and Culture College, Beijing, China; (2) Beijing Philosophy and Social Science Research Center for Intercultural Studies, Beijing Language and Culture University, Beijing, China
Corresponding author: Fan, Juanjuan(fanjuanjuan@bjhwxy.com)
Source title: Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
Abbreviated source title: Lect. Notes Comput. Sci.
Volume: 14514 LNAI
Part number: 1 of 2
Issue title: Chinese Lexical Semantics - 24th Workshop, CLSW 2023, Revised Selected Papers
Issue date: 2024
Publication year: 2024
Pages: 239-253
Language: English
ISSN: 03029743
E-ISSN: 16113349
ISBN-13: 9789819705825

Document type: Conference article (CA)
Conference name: 24th Workshop on Chinese Lexical Semantics, CLSW 2023
Conference date: May 19, 2023 - May 21, 2023
Conference location: Singapore, Singapore
Conference code: 308849
Publisher: Springer Science and Business Media Deutschland GmbH
Abstract: Based on the theory of sentiment analysis and semantic prosody, this paper investigates the semantic shift of uncivilized words in bullet-screen comments by analyzing the sentiment polarity of co-occurrence emotional words. According to the swear level and frequency of uncivilized words in the bullet-screen dataset, six kinds of words are selected for analysis. Within both the bullet-screen dataset and the BCC corpus, the semantic shifts of prototypes and their alternative forms are gauged using the sentiment lexicon and the PMI-IR algorithm. The paper offers a statistical analysis of the semantic shifts observed in these uncivilized terms and subsequently delves into the underlying motivations for these shifts.
© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2024.
Number of references: 22
Main heading: Semantics
Controlled terms: Sentiment analysis
Uncontrolled terms: Bullet-screen comment - Co-occurrence - Emotional words - PMI-IR algorithm - Semantic shift - Sentiment analysis - Sentiment lexicons - Uncivilized word
Classification code: 723.2 Data Processing and Image Processing
DOI: 10.1007/978-981-97-0583-2_19
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

5. Research on Emotion Classification Method of Bullet Screen Text Based on Video Topic Discovery

Accession number: 20251518228163
Authors: Han, Songlin (1); Gan, Gang (1)
Author affiliation: (1) Chengdu University of Information Technology, Chengdu, China
Corresponding author: Gan, Gang(test_me@cuit.edu.cn)
Source title: Proceedings - 2024 5th International Conference on Computer, Big Data and Artificial Intelligence, ICCBD+AI 2024
Abbreviated source title: Proc. - Int. Conf. Comput., Big Data Artif. Intell., ICCBD+AI
Part number: 1 of 1
Issue title: Proceedings - 2024 5th International Conference on Computer, Big Data and Artificial Intelligence, ICCBD+AI 2024
Issue date: 2024
Publication year: 2024
Pages: 283-288
Language: English
ISBN-13: 9798331533991
Document type: Conference article (CA)
Conference name: 5th International Conference on Computer, Big Data and Artificial Intelligence, ICCBD+AI 2024
Conference date: November 1, 2024 - November 3, 2024
Conference location: Jingdezhen, China
Conference code: 207890
Sponsor: Jingdezhen Ceramic University
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: With the rapid development of short video platforms, bullet screens (danmu), as a real-time way for users to express their emotions, contain rich emotional information and interactivity. However, the brevity and colloquial nature of bullet screen texts present certain challenges for emotion classification. To address this, this paper proposes an emotion classification method for bullet screen texts based on the BERT-BiLSTM-CNN model, combined with video topic features extracted by the improved HW-LDA model. The experimental results demonstrate that the emotion classification model, which incorporates topic features, significantly outperforms models without topic features in terms of accuracy, recall, and F1 score. This study highlights the importance of topic features in bullet screen emotion classification and provides new insights for emotion analysis of short texts. © 2024 IEEE.
Number of references: 17
Uncontrolled terms: BERT-BiLSTM-CNN - Bullet screen text - Classification methods - Emotion classification - Emotional information - HW-LDA - Real- time - Topic Discovery - Topic feature - Video-platforms
DOI: 10.1109/ICCBD-AI65562.2024.00054
Compendex references: YES

Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

6. A Study on the Effect of Teaching Effectiveness of Online Courses and the Number of Bullet Screen

Accession number: 20233814755167
Authors: Wang, Gengyi (1); Tian, Wenda (1); Qin, Qianhang (1)
Author affiliation: (1) Guangxi Normal University, Guilin; 541000, China
Corresponding author: Wang, Gengyi(353965085@qq.com)
Source title: Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
Abbreviated source title: Lect. Notes Comput. Sci.
Volume: 14033 LNCS
Part number: 4 of 5
Issue title: Design, User Experience, and Usability - 12th International Conference, DUXU 2023, Held as Part of the 25th HCI International Conference, HCII 2023, Proceedings
Issue date: 2023
Publication year: 2023
Pages: 200-211
Language: English
ISSN: 03029743
E-ISSN: 16113349
ISBN-13: 9783031357077
Document type: Conference article (CA)
Conference name: 12th International Conference Design, User Experience, and Usability Conference, DUXU 2023, held as part of the 25th International Conference on Human-Computer Interaction, HCII 2023
Conference date: July 23, 2023 - July 28, 2023
Conference location: Copenhagen, Denmark
Conference code: 297849
Publisher: Springer Science and Business Media Deutschland GmbH
Abstract: Compared with offline courses, the interaction between teachers and students, and between students and students in online courses is relatively simple and mainly realized through bullet screen. The presence of bullet screen can help guide students to participate in classroom interactions and enhance the attractiveness of the course, but also brings problems such as the increase in the perceived difficulty of knowledge points and the increase in the cognitive load of course knowledge points. In order to optimize the learning effect of online courses and promote the effective development of online courses, this paper focuses on the influence of the number of bullet screen on the teaching effect of online courses through an experimental design. In the experiment, the number of bullet screen was divided into 4 levels: online videos with no bullet screen, online videos with about 33% bullet screen, Online videos with about 66% bullet screen, Online videos with about 100% bullet screen. After the subjects watched online instructional videos with different bullet screen number levels in groups, the answer scores of different groups of subjects were analyzed. Following the experiment, this paper interviewed the subjects to understand their main concerns, self-assessed concentration level, concentration duration, and overall experience of online video watching while watching online courses. Finally, this paper draws the following conclusions: (1) The teaching effect of the online course with about 33% of bullet screen is excellent, learners get good scores and learn with low cognitive load and reasonable sight distribution. (2) The effect of video teaching without bullet screen is also not so bad, learners get good learning experience, but the time for learners to continue to focus on learning is shorter, so online teaching videos with no bullet screen are suitable for shorter online courses. (3) Online videos with about 66% bullet screen as well as online videos with about 100% bullet screen are less suitable for online teaching, and learners will have a greater cognitive load pressure and poorer learning results. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.
Number of references: 9
Main heading: Students
Controlled terms: Curricula - Design of experiments - E-learning - Teaching
Uncontrolled terms: Bullet screen - Cognitive loads - Number - Offline - Online course - Online teaching - Online video - Teachers' - Teaching effectiveness - Teaching effects
Classification code: 901.2 Education - 901.3 Engineering Research
Numerical data indexing: Percentage 1.00E+02%, Percentage 3.30E+01%, Percentage 6.60E+01%
DOI: 10.1007/978-3-031-35708-4_15
Funding Details: Number: 2021JGA33, Acronym: -, Sponsor: -; Number: -, Acronym: GXNU, Sponsor: Guangxi Normal University;

Funding text: Support Project. Education Teaching Reform Project of Guangxi Normal University. Project Name: Study on Optimization of Interactive Interface Design Course under the Background of New Liberal Arts Construction. Project No: 2021JGA33.
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

7. A method for aggregating the features of bullet screen

Accession number: 20234014823629
Authors: Chen, Zijian (1); Li, Shaoshuai (1); Yu, Wenbin (1)
Author affiliation: (1) Faculty of Infomation Technology, Beijing University of Technology, Beijing, China
Corresponding author: Chen, Zijian(932483811@qq.com)
Source title: Proceedings of SPIE - The International Society for Optical Engineering
Abbreviated source title: Proc SPIE Int Soc Opt Eng
Volume: 12719
Part number: 1 of 1
Issue title: Second International Conference on Electronic Information Technology, EIT 2023
Issue date: 2023
Publication year: 2023
Article number: 1271924
Language: English
ISSN: 0277786X
E-ISSN: 1996756X
CODEN: PSISDG
ISBN-13: 9781510666610
Document type: Conference article (CA)
Conference name: 2nd International Conference on Electronic Information Technology, EIT 2023
Conference date: March 31, 2023 - April 2, 2023
Conference location: Wuhan, China
Conference code: 192465
Sponsor: Academic Exchange Information Centre (AEIC)
Publisher: SPIE
Abstract: Bullet screen is a new type of comment appearing on video websites. When the video is played, it will be displayed in real time. Through sentimental analysis of the screen, we can quickly understand the sentimental types of the current video. However, it is unrealistic to deal with all the bullet screens of the whole video at one time with traditional methods. This paper proposes a method of aggregating all text content in video to represent very long video content to achieve video feature extraction. The method proposed in this paper is more effective than PCA and scalable methods in reducing dimension and achieving aggregation. © 2023 SPIE.
Number of references: 7
Main heading: Sentiment analysis
Controlled terms: Scalability
Uncontrolled terms: 'current - Bullet screen - One-time - Real- time - RoBERTa - Sentiment analysis - Text content - Transformer - Video contents - Video features
Classification code: 723.2 Data Processing and Image Processing - 961 Systems Science
DOI: 10.1117/12.2685628
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

8. Research on Relationship Between Bullet Screen Attributes in Medical Popular Science Videos and The audience's cognitive load

Accession number: 20240115300542
Authors: Xu, Wenjuan (1); Tong, Zheng (2)
Author affiliation: (1) Sanda University, College of Art Design and Media, Shanghai, China; (2) Shanghai Yf Vocational & Polytechnic School, Shanghai, China
Corresponding author: Xu, Wenjuan(34425965@qq.com)
Source title: Proceedings of the 2023 IEEE 6th International Conference on Knowledge Innovation and Invention, ICKII 2023

Abbreviated source title: Proc. IEEE Int. Conf. Knowl. Innov. Invent., ICKII
Part number: 1 of 1
Issue title: Proceedings of the 2023 IEEE 6th International Conference on Knowledge Innovation and Invention, ICKII 2023
Issue date: 2023
Publication year: 2023
Pages: 585-589
Language: English
ISBN-13: 9798350323535
Document type: Conference article (CA)
Conference name: 6th IEEE International Conference on Knowledge Innovation and Invention, ICKII 2023
Conference date: August 11, 2023 - August 13, 2023
Conference location: Sapporo, Japan
Conference code: 195074
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: This article aims to explore the relationship between bullet screen attributes in medical popular science videos and the audience's cognitive load. The evaluation indicators for cognitive load included subjects' blink frequency, gaze count, and response to task completion in eye-tracking experiments. The connection between the audience's cognitive load and the opacity and speed attributes of bullet screens in popular science videos was investigated, and the impact of different bullet screen attributes on the audience's cognitive load was explored. By analyzing the relationship between attributes such as bullet screen content, density, speed, and the audience's cognitive load in watching medical science videos, the attention allocation patterns of the audience towards bullet screens and video content were analyzed using eye-tracking technology. Experimental results indicated that the opacity of bullet screens impacted cognitive load significantly with the lowest cognitive load of audiences. The speed of bullet screens also affected the audience's cognitive load with slower speeds resulting in a less cognitive impact and lower cognitive load. © 2023 IEEE.
Number of references: 6
Main heading: Opacity
Controlled terms: Eye tracking
Uncontrolled terms: Allocation patterns - Blink frequencies - Bullet screen parameter - Cognitive loads - Evaluation indicators - Eye tracking technologies - Eye-tracking - Medical popular science video - Medical science - Screen parameters
Classification code: 741.1 Light/Optics
DOI: 10.1109/ICKII58656.2023.10332737
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

9. Turning Chatters into Donators: An Investigation of Topic-Based Bullet Screen Mode on a Livestreaming Platform

Accession number: 20241916073200
Authors: Pu, Xiaodie (1); Sindihebura, Tanguy Tresor (1); Guo, Yue (2); Chong, Alain Yee Loong (1)
Author affiliation: (1) Nottingham University Business School China, University of Nottingham Ningbo China, 199 Taikang Road, Ningbo, China; (2) College of Business, Southern University of Science and Technology, 1088 Xueyuan Avenue, Shenzhen, China
Source title: International Conference on Information Systems, ICIS 2023: "Rising like a Phoenix: Emerging from the Pandemic and Reshaping Human Endeavors with Digital Technologies"
Abbreviated source title: Int. Conf. Inf. Syst., ICIS: "Rising like Phoenix: Emerg. Pandemic Reshaping Hum. Endeavors Digit. Technol."
Part number: 1 of 1
Issue title: International Conference on Information Systems, ICIS 2023: "Rising like a Phoenix: Emerging from the Pandemic and Reshaping Human Endeavors with Digital Technologies"
Issue date: 2023
Publication year: 2023
Language: English
ISBN-13: 9781713893622
Document type: Conference article (CA)
Conference name: 44th International Conference on Information Systems: Rising like a Phoenix: Emerging from the Pandemic and Reshaping Human Endeavors with Digital Technologies, ICIS 2023
Conference date: December 10, 2023 - December 13, 2023

Conference location: Hyderibad, India
Conference code: 199133
Publisher: Association for Information Systems
Abstract: Despite the importance of social interaction in virtual communities, scant research has investigated the outcomes of social interaction features. Our study aims to investigate the business value of social interaction features in the context of livestreaming platforms. Specifically, we investigate the effect of the activation of topic-based bullet screen mode – an interactive feature which allows the streamers to set a theme or topic for the viewers to send bullet screen comments about. Our results from the regression discontinuity estimation suggest that the activation of the topic-based bullet screen mode yields an immediate decrease in viewers' chat interaction, which challenges the conventional wisdom social interaction features is a panacea for boosting increased user engagement. Nevertheless, we observe a compensatory effect whereby the decrease in chat interaction was accompanied by a surge in gift donations. This counterintuitive finding highlights the intricate interplay between social interaction features, user motivations, and platform affordances. © 2023 International Conference on Information Systems, ICIS 2023: "Rising like a Phoenix: Emerging from the Pandemic and Reshaping Hu. All Rights Reserved.
Number of references: 40
Main heading: Chemical activation
Controlled terms: Information systems - Information use - Virtual reality
Uncontrolled terms: Bullet screen - Business value - Danmu - Gifting - Interaction features - Interactive features - Livestreaming - Social interactions - User engagement - Virtual community
Classification code: 723 Computer Software, Data Handling and Applications - 802.2 Chemical Reactions - 804 Chemical Products - 903.2 Information Dissemination - 903.3 Information Retrieval and Use
Funding Details: Number: 72102116, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: -, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: LQ22G020011, Acronym: -, Sponsor: -;
Funding text: This study is funded by grants from the National Natural Science Foundation of China (72102116) and Zhejiang Natural Science Program (LQ22G020011).
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

10. Bullet-Screen-Emoji Attack With Temporal Difference Noise for Video Action Recognition

Accession number: 20243717030243
Authors: Zhang, Yongkang (1, 2); Zhang, Han (1); Li, Jun (1); Shi, Zhiping (1); Yang, Jian (1); Yang, Kaixin (3); Yin, Shuo (1); Liang, Qiuyan (1); Liu, Xianglong (4)
Author affiliation: (1) Capital Normal University, Information Engineering College, Beijing; 100048, China; (2) Beihang University, State Key Laboratory of Virtual Reality Technology and Systems, Beijing; 100191, China; (3) Kaixin Yang Is with University of Chinese Academy of Sciences, Beijing; 101408, China; (4) Beihang University, State Key Laboratory of Software Development Environment, Beijing; 100191, China
Corresponding author: Li, Jun(junmuzi@gmail.com)
Source title: IEEE Transactions on Circuits and Systems for Video Technology
Abbreviated source title: IEEE Trans Circuits Syst Video Technol
Volume: 35
Issue: 1
Issue date: 2025
Publication year: 2025
Pages: 589-600
Language: English
ISSN: 10518215
E-ISSN: 15582205
CODEN: ITCTEM
Document type: Journal article (JA)
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: Recent studies have shown that video action recognition models are also vulnerable to fooling by adversarial samples. However, currently existing video attack methods usually require high computational overhead (e.g., they generate adversarial perturbations for all frames by default), and most of them are difficult to implement printable attacks in the physical world. To address the above issues, we devise a novel efficient and effective framework for video action recognition attack: Bullet-Screen-Emoji Attack with Temporal Difference Noise (BSE), a reinforcement learning-based black-box attack method that fools the model by simply generating adversarial bullet screens for key frame and scrolling them on clean video. The agent is optimized to make the optimal actions, i.e., searching key frame. Moreover, we introduce a simple and effective temporal difference noise to enhance the attack capability of the adversarial bullet screen and accelerate the convergence

speed. Most importantly, BSE enables printable physical attacks. Extensive experiments show that our proposed BSE achieves promising attack performance on mainstream datasets (HMDB51, UCF101 and Kinetics-400) and in the physical world with high efficiency. © 1991-2012 IEEE.
Number of references: 87
Controlled terms: Reinforcement learning
Uncontrolled terms: Action recognition - Attack methods - Computational overheads - Key-frames - Physical world - Recognition models - Reinforcement learnings - Temporal difference noise - Temporal differences - Video adversarial attack
Classification code: 1101.2 Machine Learning
DOI: 10.1109/TCSVT.2024.3455799
Funding Details: Number: 62272324,62106158, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: KM202210028007,KZ20231002822, Acronym: -, Sponsor: Beijing Municipal Commission of Education; Number: 4222023, Acronym: -, Sponsor: -;
Funding text: This work was supported in part by the National Nature Science Foundation of China under Grant 62106158 and Grant 62272324, in part by the Research and Development Program of Beijing Municipal Education Commission under Grant KM202210028007 and Grant KZ20231002822, and in part by the Municipal Natural Science Foundation of Beijing of China under Grant 4222023.This work was supported in part by the National Nature Science Foundation of China under Grant 62106158; in part by the Research and Development Program of Beijing Municipal Education Commission under Grant KM202210028007; and in part by the R&D Program of Beijing Municipal Education Commission (KZ20231002822). (Yongkang Zhang and Han Zhang contributed equally to this work.) (Corresponding authors: Jun Li; E-mail: junmuzi@gmail.com.) Yongkang Zhang, Han Zhang, Jun Li, Zhiping Shi, Jian Yang, Shuo Yin and Qiuyan Liang are with the Information Engineering College, Capital Normal University, Beijing 100048, China.
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

11. Do Real-Time Reviews Matter? Examining how Bullet Screen Influences Consumers' Purchase Intention in Live Streaming Commerce

Accession number: 20224613111277
Authors: Zeng, Qingfeng (1); Guo, Qian (2); Zhuang, Wei (1); Zhang, Yu (1); Fan, Weiguo (3)
Author affiliation: (1) Department of Information Management and Electronic Commerce, Shanghai University of Finance and Economics, 777 Guoding Rd, Shanghai, China; (2) School of Business, Henan University, Henan, Kaifeng, China; (3) Department of Management Sciences, Tippie College of Business, University of Iowa, Iowa City; IA, United States
Corresponding author: Guo, Qian(gq2018310151@163.sufe.edu.cn)
Source title: Information Systems Frontiers
Abbreviated source title: Inf. Syst. Front.
Volume: 25
Issue: 5
Issue date: October 2023
Publication year: 2023
Pages: 2051-2067
Language: English
ISSN: 13873326
E-ISSN: 15729419
Document type: Journal article (JA)
Publisher: Springer
Abstract: Bullet screen as a type of real-time reviews published by viewers is an important unique feature in live streaming commerce. However, it is unclear whether and how bullet screen affects consumers' purchase intention. Drawing upon elaboration likelihood model, this study developed a conceptual model integrating two variables for central cues and three variables for peripheral cues to explain consumers' purchase intention. Using a big dataset of 668,591 records from the Taobao Live, the largest live streaming commerce platform in China, our empirical study finds that bullet screen has a vital role in influencing consumers' purchase intention. Interestingly, bullet screen sentiment has a curvilinear relationship with purchase intention, and source credibility has a negative effect on purchase intention. In addition, product type moderates the impact of bullet screen on purchase intention, and the results indicate that peripheral cues have a stronger influence on purchase intention of experience products than search products. Our study contributes to customer review literature by exploring the influence of real-time reviews on consumer behaviors in live streaming commerce. It also offers a useful practical framework for platform providers and sellers. © 2022, The Author(s), under exclusive licence to Springer Science +Business Media, LLC, part of Springer Nature.

Number of references: 74
Main heading: Commerce
Controlled terms: Consumer behavior - Purchasing - Sales - Sentiment analysis
Uncontrolled terms: Bullet screen - Consumer purchase - Elaboration likelihood models - Live streaming - Live streaming commerce - Purchase intention - Real- time - Real-time review - Sentiment analysis - Unique features
Classification code: 723.2 Data Processing and Image Processing - 931.3 Atomic and Molecular Physics
DOI: 10.1007/s10796-022-10356-4
Funding Details: Number: 71102138, Acronym: NSFC, Sponsor: National Natural Science Foundation of China;
Funding text: This work is supported by grants from the National Natural Science Foundation of China (71102138).
Compendex references: YES
ErratuFlg: 2020199051
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

12. Understanding Interaction Patterns of Bullet-Screen Videos: A Knowledge Building Perspective

Accession number: 20234314950069
Authors: Wong, Chingying (1); Zhang, Jingjing (1)
Author affiliation: (1) Beijing Normal University, Faculty of Education, Beijing, China
Corresponding author: Zhang, Jingjing(jingjing.zhang@bnu.edu.cn)
Source title: Proceedings - 2023 5th International Conference on Computer Science and Technologies in Education, CSTE 2023
Abbreviated source title: Proc. - Int. Conf. Comput. Sci. Technol. Educ., CSTE
Part number: 1 of 1
Issue title: Proceedings - 2023 5th International Conference on Computer Science and Technologies in Education, CSTE 2023
Issue date: 2023
Publication year: 2023
Pages: 282-287
Language: English
ISBN-13: 9798350325720
Document type: Conference article (CA)
Conference name: 5th International Conference on Computer Science and Technologies in Education, CSTE 2023
Conference date: April 21, 2023 - April 23, 2023
Conference location: Xi#an, China
Conference code: 193085
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: When innovative competence development is attached great importance, knowledge building is proved to be effective while learning. However, a CSCL setting for knowledge building is inaccessible to millions of online learners who study by asynchronous online videos. Students who opt to study independently online may be missing out on a more beneficial educational experience. However, the success of a Chinese website called Bilibili using the technic of bullet-screens may demonstrate the opposite. The technic of bullet-screens is widely used in online videos, but is seldom connected with knowledge building. This study explores the bullet-screen video interaction patterns from the perspective of knowledge building. By constructing a semantic network of all screens, we discovered that they might be separated into two communities, 'knowledge development' and 'community construction', thereby satisfying the desire for a knowledge building environment. There were a total of 10,164 bullet-screens coded to learn more about community growth and development. To analyze how knowledge develops, we focused on knowledge-related screens from a 4-minute video clip. Two additional semantic networks were constructed to mimic the data that students with and without bullet-screens would get, respectively. The findings show that topics in screens are authentic and diversified, giving bullet-screens a significant value in the expression of students' ideas. With their participation in discussions, learners engage in 'quasi-knowledge building' that is analogous to CSCL. © 2023 IEEE.
Number of references: 19
Main heading: Students
Controlled terms: Buildings - Computer aided instruction - E-learning - Semantics
Uncontrolled terms: Bullet-screen - Competence development - Educational experiences - Interaction pattern - Knowledge building - Knowledge development - Online learning - Online video - Semantics networks - Video interactions
Classification code: 402 Buildings and Towers - 723.5 Computer Applications - 901.2 Education
DOI: 10.1109/CSTE59648.2023.00056
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village

Compilation and indexing terms, Copyright 2026 Elsevier Inc.

13. PiEvE: A Dual-mode Game-based Plug-in Approach Applying Bullet Screen in Virtual Classrooms

Accession number: 20232614296785
Authors: Li, Xiang (1); Tian, Ye (1); Xu, Ling (1)
Author affiliation: (1) The Kyoto College of Graduate Studies for Informatics, Applied Information Technology, Kyoto, Japan
Source title: Proceedings - 2023 International Conference on Artificial Intelligence and Education, ICAIE 2023
Abbreviated source title: Proc. - Int. Conf. Artif. Intell. Educ., ICAIE
Part number: 1 of 1
Issue title: Proceedings - 2023 International Conference on Artificial Intelligence and Education, ICAIE 2023
Issue date: 2023
Publication year: 2023
Pages: 67-71
Language: English
ISBN-13: 9781665472906
Document type: Conference article (CA)
Conference name: 2023 International Conference on Artificial Intelligence and Education, ICAIE 2023
Conference date: March 20, 2023 - March 22, 2023
Conference location: Kobe, Japan
Conference code: 189274
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: In recent years, educational institutes all around the world are adversely affected by the epidemic with the social distancing or lock-downs. E-learning and the virtual classroom has been one of the most favorable solution to overcome the difficulties. With the aim of helping real-time interaction in the on-demand online learning environment, this paper focus on two common information interaction methods among younger generations: 1) bullet screen; 2) PvE mode, and discusses their motivational effect for the learners. Furthermore, we formulate a novel all-course-applicable plug-in approach beneficial in learn-ers' engagement and concentration. © 2023 IEEE.
Number of references: 10
Main heading: E-learning
Controlled terms: Computer aided instruction
Uncontrolled terms: Bullet screen - Distant learning - Dual modes - E - learning - Educational Institutes - Game-Based - Game-based Learning - Plug-ins - PvE mode - Virtual Classroom
Classification code: 723.5 Computer Applications - 901.2 Education
DOI: 10.1109/ICAIE56796.2023.00027
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

14. Research on Online Course Design Strategy Based on Text Analysis of Bullet Screen

Accession number: 20234515019942
Authors: Chen, Yuxi (1); Deng, Peng (1)
Author affiliation: (1) Yunnan Normal University, School of Information Science and Technology, Kunming, China
Corresponding author: Deng, Peng(yndenken@163.com)
Source title: 2023 6th International Conference on Big Data and Artificial Intelligence, BDAI 2023
Abbreviated source title: Int. Conf. Big Data Artif. Intell., BDAI
Part number: 1 of 1
Issue title: 2023 6th International Conference on Big Data and Artificial Intelligence, BDAI 2023
Issue date: 2023
Publication year: 2023
Pages: 108-112
Language: English
ISBN-13: 9798350303667
Document type: Conference article (CA)
Conference name: 6th International Conference on Big Data and Artificial Intelligence, BDAI 2023
Conference date: July 7, 2023 - July 9, 2023
Conference location: Haining, China
Conference code: 192965
Sponsor: Yangtze Delta Region Institute of Tsinghua University Zhejiang

Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: Based on the instructional design model ADDIE and the principle of goal-oriented course design, this study uses tools such as Python and GooSeeKer to analyze the high-frequency words, quantity changes and proportions, semantic relationships, and emotional changes in the bullet screen text of the online open course 'Grade 9 Mathematics' on Bilibili, using a combined quantitative and qualitative analysis method. It was found that learners can engage in creative thinking regarding course content, exhibit high levels of semantic association during interactions, and provide positive feedback on the course overall. However, there are also issues such as a lack of effective supervision leading to deviation from preset goals and fluctuating levels of learning engagement. Based on the analysis results, a model for online course design is proposed, with recommendations for online course development from four dimensions: course goals, course design principles, technical means, and learning feedback. This study attempts to answer questions such as 'how to design effective and learner-satisfactory online courses.' © 2023 IEEE.
Number of references: 11
Main heading: Curricula
Controlled terms: Association reactions - E-learning - Python - Semantics
Uncontrolled terms: Bullet screen text - Course design - Design strategies - Effective strategy - Goal-oriented - High-frequency words - Instructional design model - Online course - Semantic relationships - Text analysis
Classification code: 723.1.1 Computer Programming Languages - 802.2 Chemical Reactions - 901.2 Education
DOI: 10.1109/BDAI59165.2023.10256858
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

15. Sentiment Analysis: An ERNIE-BiLSTM Approach to Bullet Screen Comments (*Open Access*)

Accession number: 20223512630792
Authors: Hsieh, Yen-Hao (1); Zeng, Xin-Ping (2)
Author affiliation: (1) Department of Business Administration, National Formosa University, Yunlin; 632301, Taiwan; (2) Department of Information Management, Tamkang University, New Taipei City; 251301, Taiwan
Corresponding author: Hsieh, Yen-Hao(yhhsiehs@nfu.edu.tw)
Source title: Sensors
Abbreviated source title: Sensors
Volume: 22
Issue: 14
Issue date: July 2022
Publication year: 2022
Article number: 5223
Language: English
ISSN: 14248220
Document type: Journal article (JA)
Publisher: MDPI
Abstract: Sentiment analysis is one of the fields of affective computing, which detects and evaluates people's psychological states and sentiments through text analysis. It is an important application of text mining technology and is widely used to analyze comments. Bullet screen videos have become a popular way for people to interact and communicate while watching online videos. Existing studies have focused on the form, content, and function of bullet screen comments, but few have examined bullet screen comments using natural language processing. Bullet screen comments are short text messages of different lengths and ambiguous emotional information, which makes it extremely challenging in natural language processing. Hence, it is important to understand how we can use the characteristics of bullet screen comments and sentiment analysis to understand the sentiments expressed and trends in bullet screen comments. This study poses the following research question: how can one analyze the sentiments ex-pressed in bullet screen comments accurately and effectively? This study mainly proposes an ERNIE-BiLSTM approach for sentiment analysis on bullet screen comments, which provides effective and innovative thinking for the sentiment analysis of bullet screen comments. The experimental results show that the ERNIE-BiLSTM approach has a higher accuracy rate, precision rate, recall rate, and F1-score than other methods. © 2022 by the authors.
Number of references: 46
Main heading: Sentiment analysis
Uncontrolled terms: Affective Computing - BiLSTM - Bullet screen comment - ERNIE - Language processing - Natural languages - Psychological state - Sentiment analysis - Text analysis - Text-mining
Classification code: 723.2 Data Processing and Image Processing
DOI: 10.3390/s22145223
Compendex references: YES

Open Access type(s): All Open Access, Gold, Green
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

16. Aspect sentiment mining of short bullet screen comments from online TV series

Accession number: 20232214164413
Authors: Liu, Jiayue (1); Zhou, Ziyao (1, 2); Gao, Ming (1, 3); Tang, Jiafu (1); Fan, Weiguo (4)
Author affiliation: (1) School of Management Science and Engineering, Key Laboratory of Big Data Management Optimization and Decision of Liaoning Province, Dongbei University of Finance and Economics, Dalian, China; (2) Institute of Systems Engineering, Dalian University of Technology, Dalian, China; (3) Center for Post-doctoral Studies of Computer Science, Northeastern University, Shenyang, China; (4) Tippie College of Business, University of Iowa, Iowa City; IA, United States
Corresponding author: Gao, Ming(gm@dufe.edu.cn)
Source title: Journal of the Association for Information Science and Technology
Abbreviated source title: J. Assoc. Soc. Inf. Sci. Technol.
Volume: 74
Issue: 8
Issue date: August 2023
Publication year: 2023
Pages: 1026-1045
Language: English
ISSN: 23301635
E-ISSN: 23301643
Document type: Journal article (JA)
Publisher: John Wiley and Sons Inc
Abstract: Bullet screen comments (BSCs) are user-generated short comments that appear as real-time overlays on many video platforms, expressing the audience opinions and emotions about different aspects of the ongoing video. Unlike traditional long comments after a show, BSCs are often incomplete, ambiguous in context, and correlated over time. Current studies in sentiment analysis of BSCs rarely address these challenges, motivating us to develop an aspect-level sentiment analysis framework. Our framework, BSCNET, is a pre-trained language encoder-based deep neural classifier designed to enhance semantic understanding. A novel neighbor context construction method is proposed to uncover latent contextual correlation among BSCs over time, and we also incorporate semi-supervised learning to reduce labeling costs. The framework increases F1 (Macro) and accuracy by up to 10% and 10.2%, respectively. Additionally, we have developed two novel downstream tasks. The first is noisy BSCs identification, which reached F1 (Macro) and accuracy of 90.1% and 98.3%, respectively, through fine-tuning the BSCNET. The second is the prediction of future episode popularity, where the MAPE is reduced by 11%–19.0% when incorporating sentiment features. Overall, this study provides a methodology reference for aspect-level sentiment analysis of BSCs and highlights its potential for viewing experience or forthcoming content optimization. © 2023 Association for Information Science and Technology.
Number of references: 83
Main heading: Sentiment analysis
Controlled terms: Machine learning - Semantics
Uncontrolled terms: 'current - Analysis frameworks - Construction method - In contexts - Neural classifiers - Real- time - Semantics understanding - Sentiment analysis - User-generated - Video-platforms
Classification code: 723.2 Data Processing and Image Processing - 723.4 Artificial Intelligence
Numerical data indexing: Percentage 1.00E+01%, Percentage 1.02E+01%, Percentage 9.01E+01%, Percentage 9.83E+01%, Percentage 1.10E+01% to 1.90E+01%
DOI: 10.1002/asi.24800
Funding Details: Number: 2022RG17, Acronym: -, Sponsor: -; Number: 2020 - KF - 11 - 11, Acronym: -, Sponsor: -; Number: 71772033,71831003,72293563, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: -, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: -, Acronym: -, Sponsor: Natural Science Foundation of Liaoning Province; Number: PT - Y202214, Acronym: DUFE, Sponsor: Dongbei University of Finance and Economics; Number: -, Acronym: DUFE, Sponsor: Dongbei University of Finance and Economics;
Funding text: This work was supported by the National Natural Science Foundation of China (71772033, 72293563, 71831003), the Natural Science Foundation of Liaoning Province, China (Joint Funds for Key Scientific Innovation Bases, 2020 - KF - 11 - 11), the research capacity enhancement project of scientific research platform of Dongbei University of Finance and Economics (PT - Y202214), and the Dalian Scientific and Technological Talents Innovation Support Plan (2022RG17).
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

17. Cross-Platform Emotion Analysis and Hot Word Mining Method Based on Live Broadcast Room Bullet Screen

Accession number: 20252218519961
Authors: Li, NingZe (1, 2); Meng, Yucheng (1, 2)
Author affiliation: (1) Shenyang Institute of Computing Technology, Chinese Academy of Sciences, Shenyang, China; (2) University of Chinese Academy of Sciences, Shenyang, China
Source title: Proceedings of the IEEE International Conference on Computer and Communications, ICCC
Abbreviated source title: Proc. IEEE Int. Conf. Comput. Commun., ICCC
Part number: 1 of 1
Issue: 2024
Issue title: 2024 10th International Conference on Computer and Communications, ICCC 2024
Issue date: 2024
Publication year: 2024
Pages: 458-463
Language: English
E-ISSN: 28377109
Document type: Conference article (CA)
Conference name: 10th International Conference on Computer and Communications, ICCC 2024
Conference date: December 13, 2024 - December 16, 2024
Conference location: Chengdu, China
Conference code: 208014
Sponsor: IEEE; Sichuan Institute of Electronics; Sichuan Institute of Electronics, Youth Talent Work Committee
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: With the vigorous growth of the live streaming industry, the live streaming barrage data encompasses a considerable amount of valuable user emotion and topic information. This paper centers on the three live streaming platforms of Douyin, Kuaishou, and Xiaohongshu. By applying the methods of emotion analysis and hot word extraction, it conducts a comparison and analysis of the emotional tendencies and hotspots of users from different platforms in live streaming interactions. The findings reveal that users of Douyin have more positive emotions and focus on entertainment stars and fashion topics. Users of the Kuaishou platform have neutral emotions and are concerned about life services and local culture; while the emotions of users are more negative, and they are interested in skin care, beauty, and lifestyle. The cross-platform analysis approach proposed in this paper can offer data support for the optimization of live streaming content and user profiling. © 2024 IEEE.
Number of references: 6
Main heading: Emotion Recognition
Uncontrolled terms: Comparison and analysis - Cross-platform - Cross-platform comparison - Emotion analysis - Hot word mining - Live barrage - Live streaming - Mining methods - Sentiment analysis - User emotions
Classification code: 1101 Artificial Intelligence - 1106.8 Computer Vision
DOI: 10.1109/ICCC62609.2024.10941984
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

18. Research on the influence of online bullet screen interactive teaching on high school students art learning investment

Accession number: 20221211820786
Authors: Xu, Hongyan (1)
Author affiliation: (1) School of Art and Design, Hubei University of Technology, Hubei Province, Wuhan, China
Corresponding author: Xu, Hongyan(1825575908@qq.com)
Source title: ACM International Conference Proceeding Series
Abbreviated source title: ACM Int. Conf. Proc. Ser.
Part number: 1 of 1
Issue title: ICETM 2021 - Proceedings of 2021 4th International Conference on Education Technology Management
Issue date: December 17, 2021
Publication year: 2021
Pages: 131-135
Language: English

ISBN-13: 9781450385800
Document type: Conference article (CA)
Conference name: 4th International Conference on Education Technology Management, ICETM 2021
Conference date: December 17, 2021 - December 19, 2021
Conference location: Virtual, Online, Japan
Conference code: 177692
Sponsor: University of Lincoln
Publisher: Association for Computing Machinery
Abstract: With the rise of Internet media, the teaching methods of education are also developing. Through the educational model constructed by computer, Internet and education, the teaching has broken through the limitations of traditional education, and the forms of educational expression show an interactive development trend. The bullet screen interaction enhances the curriculum viscosity and learning situation, and creates a new model of distance communication and learning, Provide preconditions for collaborative cooperation and in-depth learning. This paper will explore whether the live barrage affects the investment level of senior high school students' art learning behavior through three experiments in remote interactive teaching, and analyze the influencing factors, so as to optimize various problems in remote interactive teaching and provide feasible suggestions for schools in remote interactive teaching. © 2021 ACM.
Number of references: 9
Main heading: Students
Controlled terms: E-learning - Education computing - Educational technology - Learning systems - Arts computing - Teaching
Uncontrolled terms: Bullet chat - Development trends - Educational modelling - High school students - Interactive development - Internet media - On-line education - Screen interaction - Teaching methods - Traditional educations
Classification code: 723.2 Data Processing and Image Processing - 901.2 Education
DOI: 10.1145/3510309.3510330
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

19. Research on the Influence of Uploaders' Behavior on the Effect of Bullet Screen Video Dissemination

Accession number: 20213210729839
Authors: Li, Yedan (1); Qi, Wei (1); Zhang, Xinzhe (1); Song, Meng (1)
Author affiliation: (1) Harbin Institute of Technology, No. 92, Xidazhi Street, Nangang District, Heilongjiang, Harbin, China
Corresponding author: Li, Yedan(770632577@qq.com)
Source title: Lecture Notes on Data Engineering and Communications Technologies
Abbreviated source title: Lecture. Notes. Data Eng. Commun. Tech.
Volume: 80
Issue date: 2022
Publication year: 2022
Pages: 799-806
Language: English
ISSN: 23674512
E-ISSN: 23674520
Document type: Book chapter (CH)
Publisher: Springer Science and Business Media Deutschland GmbH
Abstract: This study examines the factors of uploaders' behavior that have an impact on the spread of bullet screen video. A model framework was built and empirically tested with a sample of 9904 pieces of video data information. The analysis concludes that the category of bullet screen video affects its dissemination effect. The number of videos clicked, coins, favorites, comments, the frequency of video submissions and the type of video uploaded positively impact the spread of videos, while the number of shares and collections negatively impacts. This study also suggests that the video title and duration have no significant impact on bullet screen numbers. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.
Number of references: 14
Main heading: Information dissemination
Uncontrolled terms: Model framework - Video data
Classification code: 903.2 Information Dissemination
DOI: 10.1007/978-3-030-81007-8_91
Funding Details: Number: -, Acronym: BBSRC, Sponsor: Biotechnology and Biological Sciences Research Council; Number: -, Acronym: NERC, Sponsor: Natural Environment Research Council; Number: -, Acronym: -, Sponsor: Royal Society;

Funding text: We are very grateful to Norbert Schäffer, Frank Vassen, Gerard Rocamora, Glen Tyler, Gillian Gilbert and Dave Hodson who provided recordings and Phil Benstead, Cath Jeffs and Alex Turner for help and hospitality in the Outer Hebrides. Ken Smith and Rhys Green provided fruitful discussion on various aspects of this study. Paolo Galeotti and three anonymous referees provided comments that improved the manuscript. T.M. Peake was supported by a Biotechnology and Biological Sciences Research Council studentship with the Royal Society for the Protection of Birds. Analysis equipment was funded by the Natural Environment Research Council and the Royal Society.
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

20. Attacking Video Recognition Models with Bullet-Screen Comments (*Open Access*)

Accession number: 20230713573961
Authors: Chen, Kai (1); Wei, Zhipeng (1); Chen, Jingjing (1); Wu, Zuxuan (1); Jiang, Yu-Gang (1)
Author affiliation: (1) Shanghai Key Lab of Intelligent Information Processing, School of Computer Science, Fudan University Shanghai Collaborative Innovation Center on Intelligent Visual Computing, China
Corresponding authors: Chen, Jingjing(chenjingjing@fudan.edu.cn); Jiang, Yu-Gang(ygj@fudan.edu.cn)
Source title: Proceedings of the 36th AAAI Conference on Artificial Intelligence, AAAI 2022
Abbreviated source title: Proc. AAAI Conf. Artif. Intell., AAAI
Volume: 36
Volume title: AAAI-22 Technical Tracks 1
Part number: 1 of 11
Issue title: Proceedings of the 36th AAAI Conference on Artificial Intelligence, AAAI 2022
Issue date: June 30, 2022
Publication year: 2022
Pages: 312-320
Language: English
ISBN-10: 1577358767
ISBN-13: 9781577358763
Document type: Conference article (CA)
Conference name: 36th AAAI Conference on Artificial Intelligence, AAAI 2022
Conference date: February 22, 2022 - March 1, 2022
Conference location: Virtual, Online
Conference code: 185285
Sponsor: Association for the Advancement of Artificial Intelligence
Publisher: Association for the Advancement of Artificial Intelligence
Abstract: Recent research has demonstrated that Deep Neural Networks (DNNs) are vulnerable to adversarial patches which introduce perceptible but localized changes to the input. Nevertheless, existing approaches have focused on generating adversarial patches on images, their counterparts in videos have been less explored. Compared with images, attacking videos is much more challenging as it needs to consider not only spatial cues but also temporal cues. To close this gap, we introduce a novel adversarial attack in this paper, the bullet-screen comment (BSC) attack, which attacks video recognition models with BSCs. Specifically, adversarial BSCs are generated with a Reinforcement Learning (RL) framework, where the environment is set as the target model and the agent plays the role of selecting the position and transparency of each BSC. By continuously querying the target models and receiving feedback, the agent gradually adjusts its selection strategies in order to achieve a high fooling rate with non-overlapping BSCs. As BSCs can be regarded as a kind of meaningful patch, adding it to a clean video will not affect people's understanding of the video content, nor will arouse people's suspicion. We conduct extensive experiments to verify the effectiveness of the proposed method. On both UCF-101 and HMDB-51 datasets, our BSC attack method can achieve about 90% fooling rate when attacking three mainstream video recognition models, while only occluding Copyright © 2022, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.
Number of references: 48
Main heading: Deep neural networks
Controlled terms: Reinforcement learning
Uncontrolled terms: Learning frameworks - Localised - Recent researches - Recognition models - Reinforcement learnings - Spatial cues - Target model - Temporal cues - Video contents - Video recognition
Classification code: 461.4 Ergonomics and Human Factors Engineering - 723.4 Artificial Intelligence
Numerical data indexing: Percentage 8.00E+00%, Percentage 9.00E+01%
DOI: 10.1609/aaai.v36i1.19907
Funding Details: Number: 20PJ1401900, Acronym: -, Sponsor: -; Number: 62032006, Acronym: NSFC, Sponsor: National Natural Science Foundation of China; Number: -, Acronym: NSFC, Sponsor: National Natural Science Foundation of China;

Number: 20511101000, Acronym: STCSM, Sponsor: Science and Technology Commission of Shanghai Municipality; Number: -, Acronym: STCSM, Sponsor: Science and Technology Commission of Shanghai Municipality;
Funding text: This work was supported in part by NSFC project (#62032006), Science and Technology Commission of Shanghai Municipality Project (20511101000), and in part by Shanghai Pujiang Program (20PJ1401900).
Compendex references: YES
Open Access type(s): All Open Access, Gold
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

21. SmartBullets: A cloud-assisted bullet screen filter based on deep learning

Accession number: 20200209948
Authors: Niu, Haoran (1); Li, Jiangnan (1); Zhao, Yu (1)
Author affiliation: (1) Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, United States
Source title: arXiv
Abbreviated source title: arXiv
Issue date: May 14, 2019
Publication year: 2019
Language: English
E-ISSN: 23318422
Document type: Preprint (PP)
Repository: arXiv
Abstract: Bullet-screen is a technique that enables the website users to send real-time comment 'bullet' cross the screen. Compared with the traditional review of a video, bullet-screen provides new features of feeling expression to video watching and more iterations between video viewers. However, since all the comments from the viewers are shown on the screen publicly and simultaneously, some low-quality bullets will reduce the watching enjoyment of the users. Although the bullet-screen video websites have provided filter functions based on regular expression, bad bullets can still easily pass the filter through making a small modification. In this paper, we present SmartBullets, a user-centered bullet-screen filter based on deep learning techniques. A convolutional neural network is trained as the classifier to determine whether a bullet need to be removed according to its quality. Moreover, to increase the scalability of the filter, we employ a cloud-assisted framework by developing a backend cloud server and a frontend browser extension. The evaluation of 40 volunteers shows that SmartBullets can effectively remove the low-quality bullets and improve the overall watching experience of viewers.
Copyright © 2019, The Authors. All rights reserved.
Number of references: 27
Main heading: Websites
Controlled terms: Deep learning - Neural networks
Uncontrolled terms: Bullet-screen - Comment filtering - Danmaku - Filter function - Filter-based - Low qualities - Natural languange processing - Real- time - Regular expressions - Screen filters
Classification code: 461.4 Ergonomics and Human Factors Engineering
Compendex references: YES
Preprint ID: 1905.05925
Preprint source website: https://arxiv.org
Preprint ID type: ARXIV
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

22. Correction to: Do Real-Time Reviews Matter? Examining how Bullet Screen Influences Consumers' Purchase Intention in Live Streaming Commerce (Information Systems Frontiers, (2023), 25, 5, (2051-2067), 10.1007/s10796-022-10356-4) (*Open Access*)

Accession number: 20224713160114
Authors: Zeng, Qingfeng (1); Guo, Qian (2); Zhuang, Wei (1); Zhang, Yu (1); Fan, Weiguo (3)
Author affiliation: (1) Department of Information Management and Electronic Commerce, Shanghai University of Finance and Economics, 777 Guoding Rd, Shanghai, China; (2) School of Business, Henan University, Henan, Kaifeng, China; (3) Department of Management Sciences, Tippie College of Business, University of Iowa, Iowa City; IA, United States
Corresponding author: Guo, Qian(gq2018310151@163.sufe.edu.cn)
Source title: Information Systems Frontiers

Abbreviated source title: Inf. Syst. Front.
Volume: 25
Issue: 5
Issue date: October 2023
Publication year: 2023
Pages: 2069
Language: English
ISSN: 13873326
E-ISSN: 15729419
Document type: Erratum (ER)
Publisher: Springer
Abstract: The email address of the third author Wei Zhuang is incorrect. The correct email address should be stated as "zw2019310184@163.sufe.edu.cn". © 2022, Springer Science+Business Media, LLC, part of Springer Nature.
DOI: 10.1007/s10796-022-10361-7
ErratuFlg: 2020079582
Open Access type(s): All Open Access, Bronze
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

23. Design of an Interactive Classroom with Bullet Screen Function in University Teaching

Accession number: 20212210419567
Authors: Yang, Rui (1); Zhou, Cun (2); Huang, Mengjie (3); Wen, Huiqing (4); Liang, Hai-Ning (5)
Author affiliation: (1) Xi'an Jiaotong-Liverpool University, Department of Intelligent Science, Suzhou; 215123, China; (2) School of Advanced Technology, Xi'an Jiaotong-Liverpool University, Suzhou; 215123, China; (3) Xi'an Jiaotong-Liverpool University, Department of Industrial Design, Suzhou; 215123, China; (4) Xi'an Jiaotong-Liverpool University, Department of Electrical and Electronic Engineering, Suzhou; 215123, China; (5) Xi'an Jiaotong-Liverpool University, Department of Computing, Suzhou; 215123, China
Source title: 2021 9th International Conference on Information and Education Technology, ICIET 2021
Abbreviated source title: Int. Conf. Inf. Educ. Technol., ICIET
Part number: 1 of 1
Issue title: 2021 9th International Conference on Information and Education Technology, ICIET 2021
Issue date: March 27, 2021
Publication year: 2021
Pages: 47-51
Article number: 9419627
Language: English
ISBN-13: 9781665419338
Document type: Conference article (CA)
Conference name: 9th International Conference on Information and Education Technology, ICIET 2021
Conference date: March 27, 2021 - March 29, 2021
Conference location: Okayama, Japan
Conference code: 168857
Sponsor: IEEE; International Academy of Computing Technology; Okayama University; South China Normal University
Publisher: Institute of Electrical and Electronics Engineers Inc.
Abstract: Classroom interaction is an important part of teaching which can greatly enhance the learning performance of students. However, the following problems still exist in conducting interactive teaching in conventional classroom, such as limited interaction format, lack of enthusiasm and unequal interaction frequency. As a widely adopted interactive tool in online video websites, the bullet screen is instant and convenient with high interactivity, which is extremely popular among the young generation such as university undergraduate students. In this paper, a software platform is designed to collect the bullet screen messages generated by students in classroom and display them instantly together with teaching materials such as lecture slides via projector, which can be later collected for teaching and learning performance analysis. This paper aims to propose a possible solution of the mentioned problems to construct a convenient and efficient interactive classroom to improve the teaching quality. © 2021 IEEE.
Number of references: 13
Main heading: Students
Controlled terms: Educational technology - Teaching
Uncontrolled terms: Classroom interaction - Interaction frequencies - Interactive classroom - Learning performance - Teaching and learning - Teaching materials - Undergraduate students - University teaching
Classification code: 901.2 Education

DOI: 10.1109/ICIET51873.2021.9419627
Funding Details: Number: TDF19/20-R19-123, Acronym: -, Sponsor: -;
Funding text: This work is supported by Teaching Development Fund of XJTLU (TDF19/20-R19-123).
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

24. Attacking Video Recognition Models with Bullet-Screen Comments

Accession number: 20210384106
Authors: Chen, Kai (1, 2); Wei, Zhipeng (1, 2); Chen, Jingjing (1, 2); Wu, Zuxuan (1, 2); Jiang, Yu-Gang (1, 2)
Author affiliation: (1) Shanghai Key Lab of Intelligent Information Processing, School of Computer Science, Fudan University, China; (2) Shanghai Collaborative Innovation, Center on Intelligent Visual Computing, China
Corresponding authors: Chen, Jingjing(chenjingjing@fudan.edu.cn); Jiang, Yu-Gang(ygj@fudan.edu.cn)
Source title: arXiv
Abbreviated source title: arXiv
Issue date: October 29, 2021
Publication year: 2021
Language: English
E-ISSN: 23318422
Document type: Preprint (PP)
Repository: arXiv
Abstract: Recent research has demonstrated that Deep Neural Networks (DNNs) are vulnerable to adversarial patches which introduce perceptible but localized changes to the input. Nevertheless, existing approaches have focused on generating adversarial patches on images, their counterparts in videos have been less explored. Compared with images, attacking videos is much more challenging as it needs to consider not only spatial cues but also temporal cues. To close this gap, we introduce a novel adversarial attack in this paper, the bullet-screen comment (BSC) attack, which attacks video recognition models with BSCs. Specifically, adversarial BSCs are generated with a Reinforcement Learning (RL) framework, where the environment is set as the target model and the agent plays the role of selecting the position and transparency of each BSC. By continuously querying the target models and receiving feedback, the agent gradually adjusts its selection strategies in order to achieve a high fooling rate with non-overlapping BSCs. As BSCs can be regarded as a kind of meaningful patch, adding it to a clean video will not affect people's understanding of the video content, nor will arouse people's suspicion. We conduct extensive experiments to verify the effectiveness of the proposed method. On both UCF-101 and HMDB-51 datasets, our BSC attack method can achieve about 90% fooling rate when attacking three mainstream video recognition models, while only occluding © 2021, CC BY.
Number of references: 49
Main heading: Deep neural networks
Controlled terms: Reinforcement learning
Uncontrolled terms: Learning frameworks - Localised - Recent researches - Recognition models - Reinforcement learnings - Spatial cues - Target model - Temporal cues - Video contents - Video recognition
Classification code: 461.4 Ergonomics and Human Factors Engineering - 723.4 Artificial Intelligence
Numerical data indexing: Percentage 8.00E+00%, Percentage 9.00E+01%
Compendex references: YES
Preprint ID: 2110.15629v2
Preprint source website: https://arxiv.org
Preprint ID type: ARXIV
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.

25. Movie summarization using bullet screen comments

Accession number: 20172003675918
Authors: Sun, Shan (1); Wang, Feng (1); He, Liang (1)
Author affiliation: (1) Shanghai Key Laboratory of Multidimensional Information Processing, Dept. of Computer Science and Technology, East China Normal University, 3663 N. Zhongshan Rd., Shanghai, China
Corresponding author: Wang, Feng(fwang@cs.ecnu.edu.cn)
Source title: Multimedia Tools and Applications
Abbreviated source title: Multimedia Tools Appl
Volume: 77

Issue: 7
Issue date: April 1, 2018
Publication year: 2018
Pages: 9093-9110
Language: English
ISSN: 13807501
E-ISSN: 15737721
CODEN: MTAPFB
Document type: Journal article (JA)
Publisher: Springer Science and Business Media, LLC
Abstract: Automatic movie summarization helps users to skim a movie in an efficient way. However, it is challenging because it requires the computer to automatically understand the movie content and the users' opinions. Most previous works rely on the movie data itself without considering the opinions of the audience. In this paper, a novel approach for automatic movie summarization is presented by exploring a new type of user-generated data, i.e. bullet screen comments, which allow the audience to comment on the movie in a real-time manner. The number of the comments on a movie segment shows the exciting degree of the audience, while the content of the comments includes the concepts (e.g. the characters and the scenes) that interest the audience. In our approach, given a movie, bullet screen comments are utilized to select candidate highlight segments which are the most commented. Then the candidates are scored based on the number and the content of the bullet screen comments. Visual diversity is also considered in the scoring process. Finally, a subset of candidates which achieves the highest score is selected to compose a summary. Our experiments carried out on movies of different genres have shown the effectiveness of our proposed approach. © 2017, Springer Science+Business Media New York.
Number of references: 38
Main heading: Motion pictures
Controlled terms: Multimedia systems - Software engineering
Uncontrolled terms: Bullet screen comments - Movie summarization - Multimedia contents - Real time - User-generated
Classification code: 723.1 Computer Programming - 723.5 Computer Applications
DOI: 10.1007/s11042-017-4807-6
Compendex references: YES
Database: Compendex
Data Provider: Engineering Village
Compilation and indexing terms, Copyright 2026 Elsevier Inc.